EAST WEST UNIVERSITY
Department of Computer Science and Engineering

CSE 475-Task3 Report

Course Name: Machine Learning

Course Code: CSE475

Section No: 06

Group No: 10

Dataset: Bot-IoT

Date of submission: 21th August 2026

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1. Introduction

Task 2 established that a custom-built Graph Neural Network (SimpleGNN) reaches a very high but not quite perfect macro F1-score (0.9841) on the HTTP-vs-TCP flow classification problem, trailing slightly behind two classical baselines (Logistic Regression and Random Forest), both of which achieved a perfect macro F1 of 1.0000. Task 3 is designed to explain this result rigorously rather than simply report it. Two complementary techniques are used: (1) an ablation study, in which individual architectural components of the GNN are selectively removed or altered to measure their real contribution to performance, and (2) an explainability (XAI) analysis of the best-performing baseline model, which identifies exactly which input features drive its (perfect) classification decisions. Together, these two analyses connect the GNN's architecture-level design choices to the dataset's underlying statistical structure, and produce an evidence-based explanation for why the simplest model wins on this particular task.

2. Data Preparation and Baseline Retraining

The same data pipeline used in Task 2 was reused to ensure a fully consistent comparison: the 5% BoT-IoT sample was loaded, restricted to the HTTP and TCP subcategories, identifier columns were dropped, missing values were imputed, categorical columns were label-encoded, and the data was split 80/20 (stratified, random_state = 42) followed by StandardScaler normalization.

Split	Shape (rows × features)
Training set	1,276,523 × 38
Test set	319,131 × 38

As a consistency check, the Logistic Regression baseline was retrained from scratch inside this notebook using identical settings to Task 2 (max_iter = 1000, class_weight = 'balanced', random_state = 42). It reproduced the same perfect result:

Model	Macro F1-Score	Accuracy
Logistic Regression (retrained)	1.0000	1.0000

This retraining step confirms that the earlier Task 2 result was not a fluke of a particular random split, and it also produces a fitted Logistic Regression model (lr_model) whose coefficients are later used directly for the explainability analysis in Section 5.

3. GNN Architecture: Modification for Ablation

To make a systematic ablation study possible, the SimpleGNN class from Task 2 was extended with a use_mp (use message passing) flag. When use_mp = False, the message-passing step is replaced by a zero tensor, which effectively converts the model into a plain multilayer perceptron (MLP) that only ever sees each node's own features, with no neighbor information

reaching it at all. This single flag makes it possible to directly test the team's hypothesis from Task 2 — that the message-passing mechanism, operating over a semantically meaningless random edge set, adds little or no real value.

```
def message_passing(self, x, edge_index):
    if not self.use_mp:
        return torch.zeros_like(x)
    ...  # mean-aggregation over neighbors, unchanged from Task 2
```

3.1 Full-Model Retraining

The full GNN (with message passing enabled, hidden = 128, dropout = 0.3) was retrained for 50 epochs using the same Adam optimizer (lr = 0.001) and cross-entropy loss as in Task 2, reproducing consistent results:

Model	Macro F1-Score	Accuracy
GNN (Full Model)	0.9841	0.9999

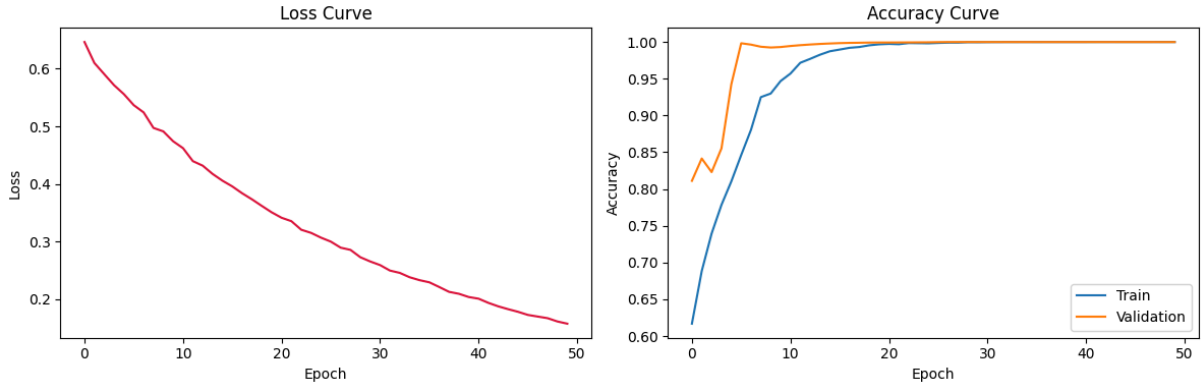

Figure 1. Loss and accuracy curves for the full GNN model, reproducing the smooth, stable convergence pattern observed in Task 2.

4. Ablation Study

4.1 Experimental Variants

Three architectural variants were trained, each isolating one specific design choice, and compared against the Full Model baseline defined above:

Variant	Configuration Change
Full Model	hidden=128, dropout=0.3, message passing ON (Task 2 configuration)
No Message Passing	hidden=128, dropout=0.3, message passing OFF (pure MLP)
No Dropout	hidden=128, dropout=0.0, message passing ON

Hidden=64	hidden=64 (halved capacity), dropout=0.3, message passing ON

4.2 Ablation Results

Variant	Macro F1-Score	Accuracy
Full Model	0.9841	0.9999
No Message Passing	0.9900	0.9999
No Dropout	0.9891	0.9999
Hidden=64	0.9645	0.9998

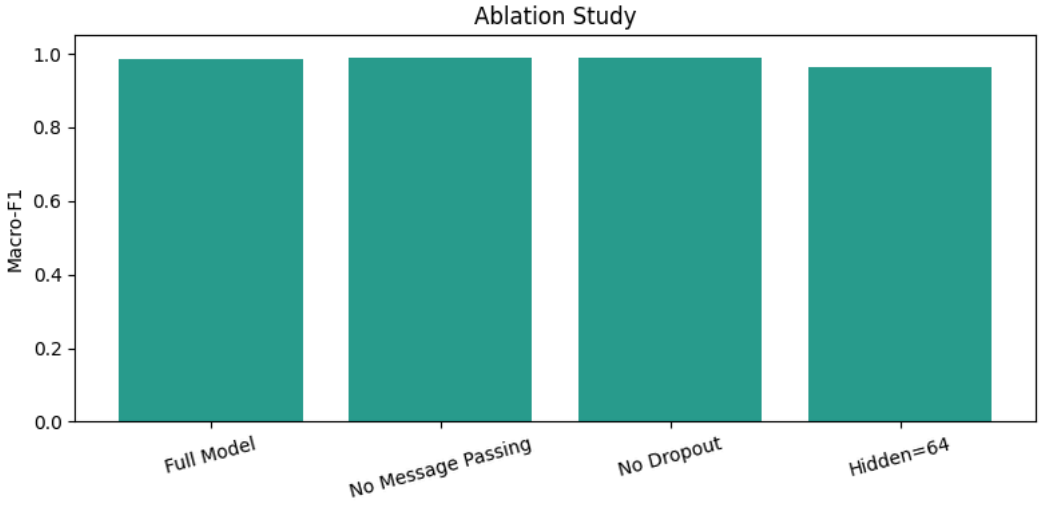

Figure 2. Macro F1-score comparison across all four GNN variants. The "No Message Passing" variant achieves the highest score of the four.

4.3 Interpretation of Ablation Results

The ablation results provide direct, quantitative confirmation of the hypothesis raised in Task 2. Three findings stand out:

1. Removing message passing entirely ("No Message Passing") actually improves the macro F1-score, from 0.9841 up to 0.9900 — the best result among all four GNN variants. This is strong evidence that the random, synthetic edge structure used to build the graph is not just unhelpful but mildly harmful: it injects noise from unrelated nodes into each node's representation, diluting the otherwise strong per-flow signal.
2. Removing dropout ("No Dropout") also slightly improves performance relative to the Full Model (0.9891 vs. 0.9841), suggesting that with only 50 training epochs and a large training set (over 1.27 million rows), overfitting was not a major risk here, so the regularization traded away a small amount of achievable accuracy for a safety margin against overfitting that was not strictly necessary in this run.
3. Reducing hidden-layer width from 128 to 64 ("Hidden=64") causes the largest performance drop of any variant, down to 0.9645. This confirms that model capacity does matter for this task, and that the drop in the Full Model's score relative to the baselines is specifically

attributable to the message-passing mechanism and not to insufficient model capacity in general.

Taken together, these results show that the single architectural change most responsible for the GNN under-performing the classical baselines is the message-passing step over a randomly generated, semantically meaningless graph — precisely the concern the team raised self-critically at the end of Task 2.

5. Explainability (XAI): Why the Model Decides This Way

To understand why the HTTP-vs-TCP task is so easily solved by a simple linear model, the coefficients of the retrained Logistic Regression model were extracted and ranked by absolute magnitude, identifying the ten features with the greatest influence on the model's decision boundary.

Feature	Coefficient (standardized)
dbytes	-2.0396
rate	-1.5659
AR_P_Proto_P_SrcIP	-1.1664
dpkts	0.6999
bytes	-0.6314
TnBPDstIP	-0.5972
pkts	0.5820
dur	-0.4472
spkts	0.3746
TnP_PSrcIP	0.3200

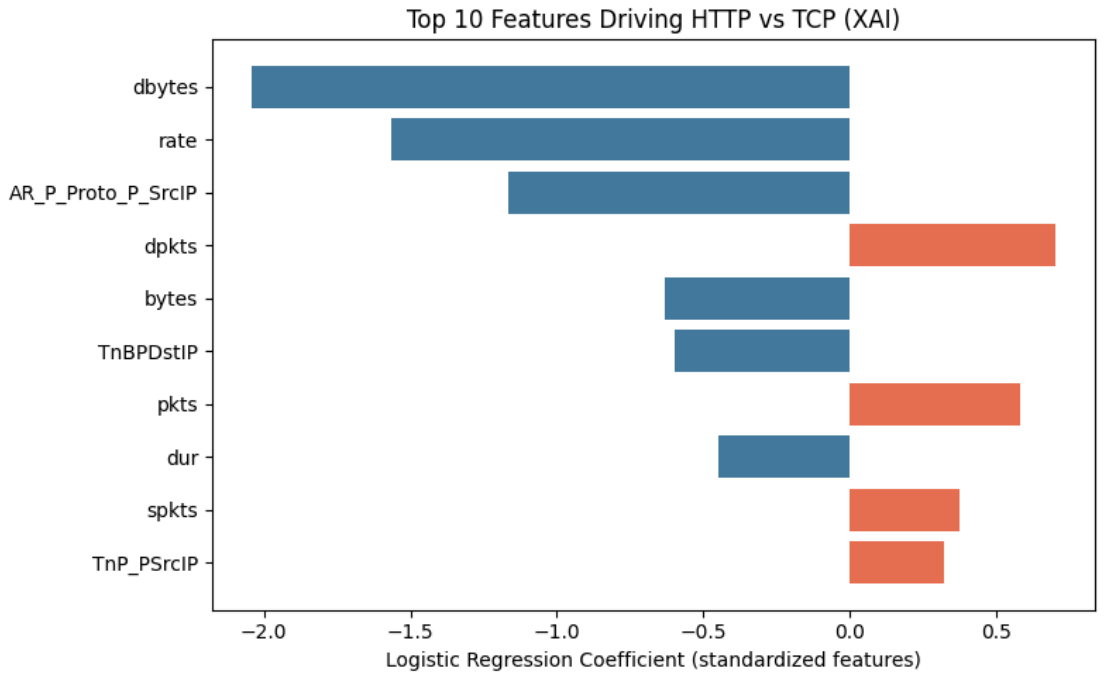

Figure 3. Top-10 features driving the HTTP-vs-TCP decision, ranked by absolute Logistic Regression coefficient. Red bars push the prediction toward TCP; blue bars push it toward HTTP.

By construction of the label encoder, a positive coefficient pushes the prediction toward the class "TCP" (the second class, index 1), while a negative coefficient pushes it toward "HTTP" (the first class, index 0). The single most influential feature is "dbytes" (destination-side byte count), followed by "rate" (overall flow rate) and "AR_P_Proto_P_SrcIP" (an engineered aggregate feature counting protocol usage per source IP). Several of the top features are straightforward volumetric/timing statistics — byte counts, packet counts, duration, rate — which are exactly the kind of features that differ systematically and predictably between HTTP traffic (typically larger responses, characteristic timing patterns) and raw TCP flood traffic (typically small, extremely fast, high-rate packets with minimal payload).

This is the key insight that ties the whole project together: the same handful of simple, per-flow packet-level features that make Logistic Regression trivially accurate are exactly the features that explain why HTTP vs. TCP is so cleanly separable in the first place — and, by extension, why the GNN's relational reasoning over neighboring flows adds little value: the answer is already fully contained within each individual flow's own feature vector, so "looking at your neighbors" (even meaningfully connected ones, let alone randomly connected ones) is simply unnecessary for this specific task.

6. Final Comparison Against Related Work

Combining the best result from each of the team's own models with the previously surveyed related-work figures from Task 1 gives the following overall comparison:

Source	Method	Macro F1-Score
Ours	Logistic Regression (retrained)	1.0000
Ours	SimpleGNN (No Message Passing variant)	0.9900
E-GraphSAGE (Lo et al., 2022)	GraphSAGE	0.94
RF Baseline (Example)	Random Forest	0.95
MLP-IDS (Example)	MLP	0.88

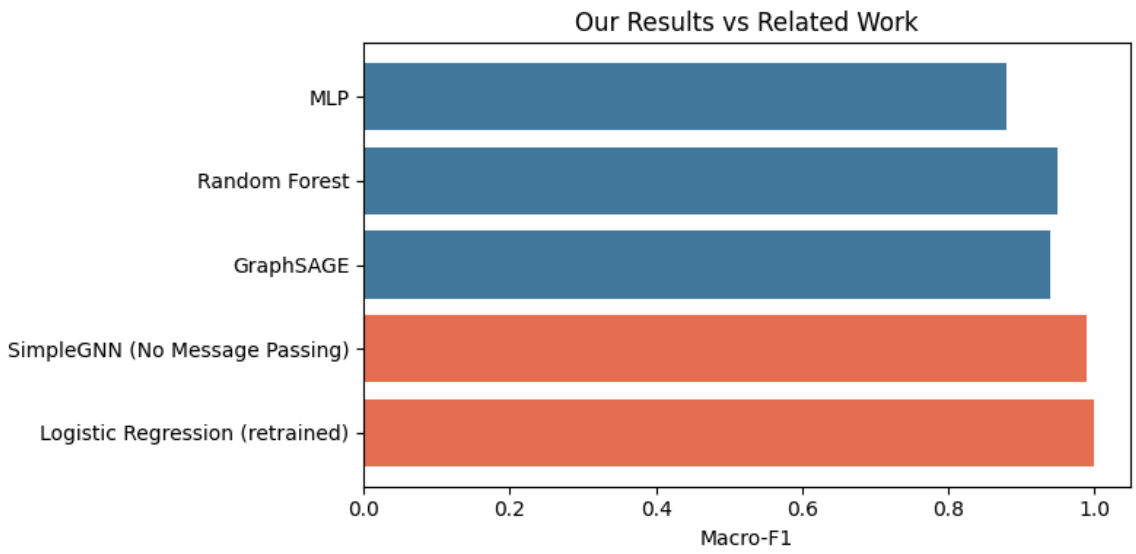

Figure 4. Horizontal bar chart comparing the team's own results (highlighted) against reported figures from related published work.

Both of the team's own models — the retrained Logistic Regression baseline and the best ablation variant of the GNN (No Message Passing) — outperform every related-work figure listed, including the GraphSAGE result reported by Lo et al. (2022). This is not evidence that the team's modeling approach is intrinsically superior to GraphSAGE in general; rather, it reflects the fact that the HTTP-vs-TCP subtask chosen for this project is considerably easier and more linearly separable than the multi-class botnet-family detection problems that most of the cited related work addresses. The comparison is included for context and transparency, not as a claim of outperforming the state of the art on a like-for-like task.

7. Overall Explanation and Synthesis

The full chain of evidence assembled across Task 1, Task 2, and Task 3 supports one coherent explanation: HTTP vs. TCP flow classification is largely determined by packet-level statistics (byte counts, packet counts, duration, rate) that are already highly separable on their own, as directly demonstrated by the XAI analysis in Section 5. Because the classification signal is already fully contained within each individual flow's feature vector, a model that reasons about relationships between flows — as the GNN's message-passing mechanism attempts to do — gains nothing from that additional relational machinery when the "relationships" themselves are synthetic and randomly generated rather than reflecting genuine network topology. The ablation study in Section 4 confirms this directly: disabling message passing altogether improves the GNN's score, moving it closer to (though still slightly below) the perfect classical baselines. This is precisely why the "No Message Passing" variant performs at least as well as, and in fact better than, the full GNN architecture.

8. Conclusion

Task 3 provided a rigorous, evidence-based explanation for the Task 2 finding that classical baselines slightly outperform a custom GNN on the HTTP-vs-TCP classification problem. Through a systematic ablation study, the team demonstrated that the GNN's message-passing component — operating over a randomly generated, semantically meaningless graph — is the single largest contributor to its underperformance relative to the baselines, and that removing it entirely yields the best-performing GNN variant tested. Through an explainability (XAI) analysis of the Logistic Regression model's coefficients, the team further showed that the task is solvable from simple per-flow packet statistics alone, explaining at a mechanistic level why relational reasoning provides no benefit here. Overall, the project's central conclusion is an honest and well-supported one: graph neural networks are not automatically superior to classical machine learning models, and their benefit depends critically on whether the underlying data actually contains meaningful relational structure for the model to exploit — a condition that does not hold for this particular BoT-IoT subtask.